Himalaya Ashish

Senior AI Developer & ML Engineer

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Dammam, Saudi Arabia

Summary

AI Developer with **13+ years of experience** specializing in **Agentic AI systems** and **Scientific Machine Learning**. Expert in building AI Agents, Advanced RAG, and Multi-Agent architectures using LangChain and LangGraph. Proven expertise in Scientific Computing, developing **PINNs** and **Neural Operators (PINO/FNO)** for reservoir simulations with parallel training and HPC deployment using MPI. Specialized in designing GPU-based AI compute environments, computer vision applications, scalable ML platforms, containerized deployments, and production-grade AI systems. Strong expertise in architecting AI workloads and building CI/CD pipelines for ML models.

Experience

Saudi Aramco

Jul 2025 – Present

AI Developer — Dammam, Saudi Arabia

- Designed and deployed production-ready **Agentic AI systems** using LangChain and LangGraph, implementing ReAct agents, Multi-Agent orchestration, and RAG pipelines for enterprise knowledge management — improving query resolution accuracy by **35%** and reducing manual effort by **40%**.
- Architected **Model Context Protocol (MCP)** integration and context-engineered memory systems (short-term, long-term, episodic) for conversational AI agents, boosting CSAT scores by **30%** and reducing average response latency by **25%** through LangSmith-monitored evaluation pipelines.
- Delivered multimodal LLM solutions including Text2SQL agents, RAG-powered QA systems, and invoice processing automation across healthcare, banking, and financial domains — reducing document processing time by **40%** and improving data extraction accuracy by **30%**.
- Developed **Physics-Informed Neural Networks (PINNs)** and Neural Operators (PINO/FNO) for reservoir simulation, embedding governing physical equations as training constraints to deliver fast, physics-consistent surrogate models.
- Executed end-to-end ML pipeline for scientific deep learning — from large-scale data extraction to distributed parallel training using **MPI** across multi-node HPC infrastructure with QRSH job scheduling, validated against physics-based simulation baselines.

IBM Labs

Jun 2023 – Jun 2025

AI Engineer — Bengaluru, India

- Built and deployed **production-grade agentic systems** using WatsonX LLMs, LangChain, and ChromaDB, integrating RAG pipelines and vector databases to improve CSAT by **30%**.
- Developed Python microservices for LLM inference, retrieval, orchestration, and monitoring; built embedding pipelines and semantic search services for structured and unstructured enterprise data.
- Implemented **CI/CD pipelines** using Docker, Kubernetes, Jenkins, and AWS, reducing deployment time by **40%** while ensuring data quality, reproducibility, and model monitoring across AI systems.
- Built scalable AI pipelines combining computer vision models with cloud-native MLOps; implemented context and knowledge management layers grounding AI outputs in trusted enterprise data.
- Architected and deployed **AWS SageMaker**-based ML workflows for real-time inference and batch prediction, integrating S3, Lambda, and CloudWatch to enable end-to-end model lifecycle management for enterprise clients.
- Leveraged **AWS Bedrock** to prototype and evaluate foundation model APIs for enterprise chatbot use cases, enabling rapid LLM selection, cost benchmarking, and production-readiness assessments.

Accolite Digital

May 2022 – Jun 2023

Senior Machine Vision Engineer — Bengaluru, India

- Delivered high-accuracy computer vision models (Detectron2, Mask R-CNN, RetinaNet, UNet, YOLOv5/v8), achieving **95% accuracy** on DICOM lumbar spine detection and **90% precision** on X-ray instrument recognition.
- Reduced image noise and false positives by **30%/20%** using autoencoders and CLAHE; enhanced industrial defect detection via MVTec Halcon achieving **98% scratch accuracy** with optimized lighting configurations.
- Designed real-time **computer vision inference pipelines** using OpenCV and TensorRT, deploying optimized models on edge devices with **3× throughput improvement** over baseline CPU inference.
- Built end-to-end CV pipelines with OpenCV, PyTorch, and TensorFlow; streamlined MLOps using Jenkins, Docker, Kubernetes, and AWS, reducing deployment time by **40%**.

Deloitte *May 2019 – May 2022*

Machine Learning Engineer Bengaluru, India

- Trained and fine-tuned **Mask R-CNN, MobileNet, ResNet, EfficientNet, and YOLOv3** for detection, segmentation, and classification across automotive and manufacturing use cases.
- Deployed MVTec Halcon for OCR/barcode extraction in automotive paint workflows, automating chassis text detection and digitizing quality control processes.
- Built **AWS CI/CD pipelines** (Git, ECR, Jenkins, Docker, Kubernetes) for seamless model deployment; implemented data governance frameworks ensuring compliance, security, and quality. *Promoted to Manager (2022).*

eClerx *Jun 2017 – May 2019*

Data Scientist Mumbai, India

- Built hybrid OCR pipeline (object detection + OCR) for oil & gas PDFs; developed NER models using Word2Vec, TF-IDF, and BERT for accurate entity extraction, improving digitization accuracy significantly.
- Created LSTM-based document summarizers and text classification models; implemented ensemble methods (stacking, boosting) improving forecasting accuracy by **25%**.

CGI *Feb 2012 – Jun 2017*

Associate Software Engineer India

- Built data pipelines using Python, Hadoop, SQL, and Hive to process **1TB+ data daily**; implemented SQL Server log-shipping and mirroring to ensure **99.9% data availability**.
- Optimized queries reducing response time by **30%**; performed P2V database migration and developed automated alert/notification scripts.

Technical Skills

Languages: **Python**, C++, **SQL**, Hive

GenAI & Agentic Ecosystem: **LangChain**, **LangGraph**, **AI Agents (ReAct, Planning)**, **Multi-Agent Orchestration**, **Model Context Protocol (MCP)**, **Advanced RAG**, LangSmith, **ChromaDB**, **FAISS**, **Pinecone**, **LoRA/PEFT**, DSPy, **LlamaIndex**, **AutoGen**, **CrewAI**

Deep Learning & Scientific ML: **PyTorch**, TensorFlow, PINNs, PINO/FNO Neural Operators, **CUDA**, **ONNX**, UNet, **YOLO**, EfficientNet, Object Detection, Image Segmentation, **OpenCV**

Backend & Engineering: **FastAPI**, Flask, **REST APIs**, **Docker**, **Kubernetes**, **CI/CD**, Webhooks, SHAP/LIME (XAI), Hallucination Detection, Model Benchmarking (BLEU/ROUGE/TruthfulQA)

Cloud & MLOps: **AWS (Bedrock, SageMaker, S3, Lambda, CloudWatch)**, **Azure OpenAI** (Prompt Flow, Cognitive Search, CosmosDB), **GCP (Vertex AI**, Model Garden), **MLflow**, Feature Store, Model Registry, Drift Detection

Data Engineering: Structured/Unstructured Data Ingestion, **Vector Databases**, **SQL/NoSQL**, HDF5, **ETL/ELT Pipelines**

Governance & Compliance: ISO 55000 , OSDU ,HIPAA, FDA 21 CFR Part 11 ,Model Explainability, Fairness Audits, GDPR, ISO 42001, NIST AI RMF

Domains: Oil & Gas (Reservoir Simulation), Banking & Finance, Healthcare, Manufacturing, Semiconductor

Publication

- **AI Accelerated Forward Modeling Using Physics-Informed Neural Operators for Real-Time Reservoir Management.** *International Petroleum Technology Conference (IPTC), 2026.*

Certifications

- **Building & Evaluating Agentic AI Systems**
- **Generative AI for Software Development**
- **TensorFlow Developer Certification**
- **Generative AI with LLM – AWS**
- **NVIDIA Deep Learning**
- **Halcon MVTec Deep Learning**

Education

Woolf University | M.S. in Artificial Intelligence & Machine Learning *Oct 2022 – Oct 2024*

GEC, Gwalior | B.E. in Information Technology *Aug 2007 – Dec 2011*

Achievements

Recognised by IBM Leadership with Blue Points Award— for outstanding contributions to AI *2024*

Promoted to Manager at Deloitte— recognizing excellence in AI and ML delivery and team leadership.*2022*

Received Pat on Back Award at Deloitte— awarded by leadership for exceptional project impact.*2021*